

Theerpu Intelligence System (TIS)

Judgment Lifecycle & Decision Intelligence for Government

Bharath Hackathon 2026

1. Introduction

Theerpu Intelligence System (TIS) is an AI-powered operational intelligence platform designed to improve government decision-making in judicial and administrative workflows.

The system focuses on:

- tracking evolving judicial instructions,
- identifying operational conflicts,
- prioritizing government actions,
- monitoring legal escalation risks,
- and providing explainable decision intelligence.

Unlike traditional legal management systems that mainly store records and documents, TIS functions as a command-center style operational intelligence platform capable of:

- lifecycle-aware case tracking,
- action prioritization,
- explainable escalation reasoning,
- and deterministic workflow simulation.

The project demonstrates how Artificial Intelligence and operational intelligence systems can modernize governance workflows and support smart judicial operations.

2. Problem Statement

Government departments and judicial stakeholders often face operational challenges while handling evolving court orders, legal instructions, and administrative actions.

Existing systems suffer from several limitations:

Existing Challenges

1. Contradictory Government Instructions

Departments may receive conflicting judicial or operational directives.

2. Manual Legal Tracking

Most workflows rely heavily on manual case monitoring and administrative review.

3. Delayed Government Actions

Critical deadlines and escalations are often identified too late.

4. Lack of Operational Prioritization

Current systems do not intelligently classify urgent government actions.

5. Poor Visibility into Case Evolution

Stakeholders struggle to track how legal instructions evolve over time.

6. Lack of Explainability

Most systems fail to explain why certain cases are flagged or escalated.

As a result, government operations become reactive instead of proactive.

3. Proposed Solution

Theerpu Intelligence System introduces an AI-driven decision intelligence platform capable of:

- monitoring judicial workflows,
- tracking instruction lifecycles,
- detecting operational conflicts,
- prioritizing legal actions,
- and providing explainable escalation reasoning.

The system transforms complex judicial operations into:

- actionable,
- explainable,
- prioritized operational intelligence.

The platform is designed as a command-center experience for government stakeholders and judicial authorities.

4. Objectives

Primary Objectives

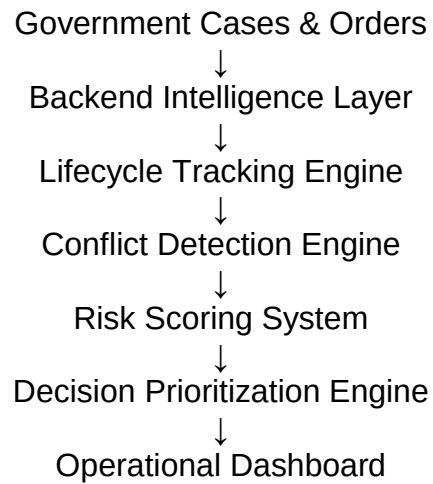
- Improve judicial workflow visibility
- Enable action-first operational intelligence
- Detect contradictory government instructions
- Prioritize urgent legal actions
- Support explainable escalation reasoning

Secondary Objectives

- Simulate realistic government workflows
- Demonstrate scalable governance intelligence
- Improve administrative efficiency
- Support future AI integrations

5. System Architecture

High-Level Architecture



6. Detailed Module Explanation

6.1 Dashboard Command Center

Description

Provides centralized operational visibility for government stakeholders.

Features

- KPI metrics
- Urgency indicators
- Lifecycle intelligence feed
- Critical conflict alerts
- Immediate action queue
- Deadline tracking
- AI interpretation summaries

Purpose

Deliver command-center style operational monitoring.

6.2 Cases Investigation Workspace

Description

Handles detailed case lifecycle analysis.

Functionalities

- Case list and details view
- Lifecycle timeline
- Instruction evolution tracking
- Connected order mapping
- Operational recommendations
- Recommendation confidence display

Objective

Provide complete operational visibility into evolving cases.

6.3 Decision Intelligence Center

Description

Prioritizes operational government actions.

Decision Categories

- Act Now
- Review Soon
- Stable

Features

- Action recommendations
- Deadline visibility
- Confidence indicators
- Risk prioritization

Goal

Enable faster and more organized decision-making.

6.4 Conflict Detection Engine

Description

Identifies contradictory instructions and operational conflicts.

Functionalities

- Before-vs-after comparison
- Conflict severity scoring
- Legal exposure identification
- Operational impact analysis

Importance

Prevents operational confusion and escalation risks.

6.5 Explainability Layer

Description

Displays WHY cases are flagged or escalated.

Explainability Includes

- escalation reasons,
- conflict severity,
- confidence levels,
- operational impact,
- lifecycle interpretation.

Importance

Improves transparency and trust in decision-making.

6.6 Demo Simulation Engine

Description

Provides deterministic operational workflow simulation.

Simulation Controls

- Inject New Judicial Order
- Simulate Deadline Escalation
- Trigger Conflict Escalation
- Officer Verification Complete
- Reset Demo Scenario

Key Innovation

The entire UI reacts consistently across:

- timelines,
- alerts,
- feeds,
- action queues,
- and operational panels.

Benefit

Provides reliable presentation-grade simulation behaviour.

7. Functional Requirements

Core Functional Requirements

Case Monitoring

The system must monitor judicial instructions and government actions.

Lifecycle Tracking

The system must track how instructions evolve over time.

Conflict Detection

The system must identify contradictory directives.

Risk Scoring

The system must prioritize legal and operational risks.

Action Prioritization

The system must classify cases into urgency levels.

Explainability

The system must explain escalation decisions clearly.

Dashboard Monitoring

The system must display operational intelligence visually.

Simulation Engine

The system must simulate operational workflow events consistently.

8. Non-Functional Requirements

Scalability

The architecture should support future government deployment.

Reliability

The system should provide deterministic and stable workflows.

Performance

Dashboard responses should remain efficient under multiple updates.

Security

Future implementations should support secure government operations.

Maintainability

The modular architecture should support future enhancements.

Usability

The dashboard should remain operationally intuitive.

9. Technology Stack

Frontend

- React
- Vite
- JavaScript
- Custom CSS

Backend

- Express.js
- REST APIs
- CORS support

Data Layer

- JSON lifecycle records
- Deterministic state updates

Intelligence Layer

- Risk scoring engine
- Conflict detection logic
- Operational prioritization system

Architecture Design

- Modular component structure
- Command-center dashboard design
- Deterministic workflow simulation

10. Workflow Explanation

Complete Workflow

Step 1 — Government Case Monitoring

Judicial instructions and operational directives are monitored.

Step 2 — Lifecycle Tracking

The system tracks evolving instructions and operational changes.

Step 3 — Conflict Analysis

Conflicting directives are identified automatically.

Step 4 — Risk Scoring

Urgency and operational exposure are calculated.

Step 5 — Decision Prioritization

Cases are classified into:

- Act Now
- Review Soon
- Stable

Step 6 — Explainability Processing

Escalation reasons and operational interpretations are generated.

Step 7 — Dashboard Visualization

Operational intelligence is displayed visually.

11. Innovation Highlights

Lifecycle-Aware Intelligence

Tracks evolving judicial instructions dynamically.

Explainable Operational Intelligence

Shows WHY cases are escalated.

Action-First Design

Focuses on operational government decisions.

Deterministic Workflow Architecture

Provides reliable and predictable behaviour.

Government Workflow Realism

Designed as a real operational intelligence command center.

12. Real-World Impact

Government Authorities

Improves operational coordination and legal monitoring.

Judicial Stakeholders

Enhances visibility into case evolution.

Administrative Systems

Supports faster operational responses.

Governance Infrastructure

Demonstrates scalable digital governance intelligence.

13. Future Scope

Planned Enhancements

- Real judicial database integration
- NLP-based order analysis
- AI-powered legal reasoning
- Multi-department deployment
- Predictive legal escalation analytics
- Smart governance integration

14. Conclusion

Theerpu Intelligence System demonstrates how AI-powered operational intelligence can modernize judicial and government workflows through:

- lifecycle-aware monitoring,
- explainable escalation,
- operational prioritization,
- and deterministic workflow intelligence.

The platform focuses on actionable government decision support instead of traditional record management systems, making it a scalable and future-ready governance intelligence solution.